EE 677 Foundation of VLSI CAD
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IIT Bombay
Assignment 2

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Assignment 2 Problem Statement:
Given a DFG, develop a program to find out schedule (list-based algorithm) and binding of this algorithm with a physical functional unit and assignment of variables with registers. Make a datapath.
Input: Data flow graph,
Output: netlist

Abstract:

High-Level Synthesis (HLS) is an important step in the digital hardware design flow that automatically transforms an algorithmic specification into a register-transfer level (RTL) implementation. The effectiveness of HLS depends on several key decisions, including operation scheduling, resource allocation, functional-unit binding, register allocation, and datapath construction.

This project presents the design and implementation of a DFG-based High-Level Synthesis framework that accepts a Data Flow Graph (DFG) as input and automatically generates a scheduled and resource-constrained hardware datapath followed by synthesizable Verilog RTL. The proposed framework represents computational operations and their dependencies using a directed acyclic graph and applies ASAP and ALAP scheduling, followed by resource-constrained list scheduling. The scheduled operations are then mapped to functional units and registers. Based on these binding decisions, the framework generates the required multiplexer and control information, constructs the datapath and netlist, and finally produces RTL code.

The generated RTL is verified using a dedicated testbench through functional simulation and waveform analysis. The project demonstrates the complete transformation from an algorithmic DFG representation to an executable RTL hardware implementation. The framework provides a structured platform for understanding and experimenting with fundamental HLS concepts such as scheduling, resource sharing, register allocation, datapath generation, and RTL generation.

Introduction

High-Level Synthesis

The increasing complexity of modern digital systems has made traditional RTL-based hardware design increasingly time-consuming. Designing every register, multiplexer, functional unit, and control signal manually requires significant effort and can make design modification and optimization difficult.

High-Level Synthesis (HLS) addresses this problem by transforming an algorithmic or behavioral specification into an RTL hardware implementation automatically. Instead of describing the hardware structure directly, the designer specifies the computation and its dependencies. The HLS system then determines an appropriate hardware architecture subject to constraints such as execution time, available resources, and storage requirements.

A typical HLS flow can be represented as:

Algorithmic Specification
↓
Intermediate Representation
↓
Scheduling
↓
Resource Allocation
↓
Functional Unit Binding
↓
Register Binding
↓
Datapath Generation
↓
RTL Generation

In this project, the Data Flow Graph (DFG) is used as the primary intermediate representation. Each computational operation is represented as a node, while data dependencies are represented as directed edges. The DFG therefore captures both the computation and the ordering constraints between operations.

The project implements the major stages required to transform this DFG into a functional RTL datapath.

Data Flow Graph (DFG)

A Data Flow Graph (DFG) is a directed graph used to represent computations and the dependencies between them. In an HLS system, the DFG provides an intermediate representation between the algorithmic description and the hardware implementation.

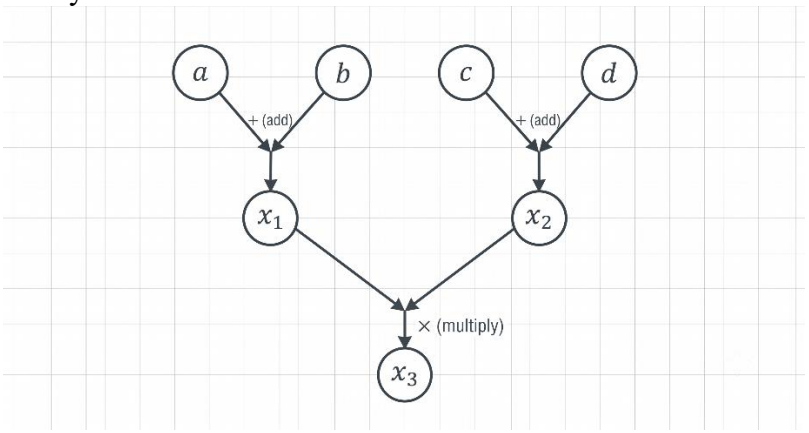
A DFG generally consists of:

1. Nodes — computational operations.
2. Edges — data dependencies.
3. Inputs — primary input operands.
4. Outputs — final computed values.

For example, consider:

$$x_1 = a + b$$
$$x_2 = c + d$$
$$x_3 = x_1 \times x_2$$

The corresponding dependency structure is:



The graph explicitly indicates that the multiplication cannot begin until both additions producing $x_1$and $x_2$have completed.

In the proposed framework, the DFG is specified using a textual input format and converted into an internal graph representation. Each node stores information such as its operation type, operands, predecessors, successors, and scheduling/binding information.

Scheduling

Scheduling determines the clock cycle in which each operation in the DFG will execute.

An HLS scheduler must satisfy two major constraints:

1. Data dependencies
2. Hardware resource availability

For example, if:

$$x_1 = a + b$$

and

$$x_2 = x_1 \times c$$

then the multiplication must occur after the addition because $x_2$depends on x_1.

ASAP Scheduling

As Soon As Possible (ASAP) scheduling assigns each operation to the earliest cycle in which all of its dependencies have been satisfied.

For an operation v:

$$ASAP(v) = \begin{cases} 1, & \text{if } v \text{ has no predecessors} \\ \max\limits_{u \in Pred(v)} \big(ASAP(u) + L(u)\big), & \text{otherwise} \end{cases}$$

where $L(u)$represents the latency of operation u.

ALAP Scheduling

As Late As Possible (ALAP) scheduling assigns each operation to the latest cycle in which it can execute without violating the overall latency constraint.
The difference between ALAP and ASAP gives the mobility:

$$M(v) = ALAP(v) - ASAP(v)$$

An operation with lower mobility has fewer scheduling alternatives.

Resource-Constrained Scheduling

After dependency analysis, the project performs resource-constrained scheduling. The scheduler considers both:

- Whether an operation is ready based on dependencies.
- Whether the required functional unit is available.

This results in a practical cycle-by-cycle schedule that respects the available hardware resources.

Resource Constraints

Hardware resources are finite. A design may contain many arithmetic operations, but it may not be desirable or possible to instantiate a separate hardware unit for every operation.
For example, suppose a DFG contains four additions but only one adder is available.
Instead of:
ADD1
ADD2
ADD3
ADD4
the HLS system can reuse: ADD1 with different operations in different cycles
This is called **resource sharing**.
The project therefore maintains resource constraints such as the number of available:

- ADD units
- SUB units
- MUL units
- DIV units

The list scheduler uses these limits when assigning operations to clock cycles.
Resource constraints create an important trade-off:

More Resources ⇒ Potentially Lower Latency

while:

Fewer Resources ⇒ Greater Resource Sharing

but potentially more execution cycles.

Functional-Unit Binding

After scheduling determines when an operation executes, functional-unit binding determines where the operation executes.
A functional unit is a hardware block capable of performing a particular class of operations, such as:

- Adder
- Subtractor
- Multiplier
- Divider

For example:

Operation	Cycle	Functional Unit
ADD	1	ADD1
ADD	2	ADD1
MUL	3	MUL1

The same ADD1 unit can therefore execute different additions in different cycles.
Functional-unit binding converts the abstract operation schedule into a hardware resource assignment.
The result of this stage is an important input to datapath generation because it determines which physical arithmetic units must be instantiated.

Register Binding

Intermediate values generated during computation must be stored until they are required by subsequent operations.
Register binding maps logical variables to physical registers.
For example:

$$x1 \rightarrow R1$$
$$x2 \rightarrow R2$$
$$x3 \rightarrow R3$$

The number of registers can be reduced when two variables have non-overlapping lifetimes.
The project therefore performs live-range analysis to determine the period during which each intermediate variable is required.
If two variables do not overlap in lifetime, the same physical register may potentially be reused.
Thus, register binding converts:

$$\text{Logical Variables} \rightarrow \text{Physical Registers}$$

and contributes directly to the storage requirements of the generated datapath.

Datapath Generation

Once scheduling and binding decisions have been completed, the project generates the actual hardware datapath.
The datapath consists primarily of:

- Functional units
- Registers
- Multiplexers
- Data connections
- Input/output interfaces

The datapath is determined by the combined results of:

$$\text{Scheduling} + \text{FU Binding} + \text{Register Binding} + \text{MUX/Control Generation}$$

The project therefore moves from an abstract computational representation to an explicit hardware architecture.

RTL Generation

Register-Transfer Level (RTL) describes digital hardware in terms of registers, combinational logic, data transfers, and clocked operations.
After the datapath and netlist have been generated, the project automatically produces Verilog RTL.
The complete transformation is:

$$\boxed{DFG \rightarrow Scheduling \rightarrow Binding \rightarrow Datapath \rightarrow Netlist \rightarrow RTL}$$

The generated RTL contains the hardware structures determined by the HLS process, including:

- Registers
- Arithmetic units
- Multiplexers
- Control logic
- Clock/reset logic

- Input/output ports

The RTL can then be simulated using a testbench to verify that the generated hardware produces the expected result.
This final stage demonstrates the principal goal of the project: automatically transforming an algorithmic DFG into a functionally verifiable RTL hardware implementation.

Proposed HLS Framework

1. input/example.dfg

```
INPUTS: a,b,c,d,e,f,g,h

NODE: N1 ADD a b x1
NODE: N2 ADD c d x2
NODE: N3 MUL x1 x2 x3
NODE: N4 ADD e f x4
NODE: N5 ADD x3 x4 x5
NODE: N6 MUL g h x6
NODE: N7 ADD x5 x6 y

OUTPUTS: y
```

This is the input description of the computational problem. It defines the primary inputs, operations, intermediate variables, and final output in a simple textual DFG format.
The example represents:

$$y = \big((a + b)(c + d)\big) + (e + f) + (g \times h)$$

It contains seven operations:

1. N1: $x1 = a + b$
2. N2: $x2 = c + d$
3. N3: $x3 = x1 \times x2$
4. N4: $x4 = e + f$
5. N5: $x5 = x3 + x4$
6. N6: $x6 = g \times h$
7. N7: $y = x5 + x6$

Purpose: Provides the computational specification that is converted into a DFG and subsequently synthesized into a hardware datapath.

2. dfg.py

```
class DFGNode:

    def __init__(self, node_id, operation, inputs, output):

        self.node_id = node_id
        self.operation = operation
        self.inputs = inputs
        self.output = output

        # Dependency information
        self.predecessors = []
```

```
        self.successors = []

        # Scheduling information
        self.asap = None
        self.alap = None
        self.mobility = None

        # Binding information
        self.functional_unit = None

    def __repr__(self):

        return (
            f"DFGNode("
            f"id={self.node_id}, "
            f"op={self.operation}, "
            f"inputs={self.inputs}, "
            f"output={self.output})"
        )

class DFG:

    def __init__(self):

        self.inputs = []
        self.nodes = []
        self.outputs = []

    def add_node(self, node):

        self.nodes.append(node)

    def get_node(self, node_id):

        for node in self.nodes:

            if node.node_id == node_id:
                return node

        return None

    def get_node_by_output(self, variable):

        for node in self.nodes:

            if node.output == variable:
                return node

        return None

    def build_dependencies(self):
```

```
        # Clear existing dependencies
        for node in self.nodes:

            node.predecessors = []
            node.successors = []

        # Determine dependencies
        for node in self.nodes:

            for input_variable in node.inputs:

                producer = self.get_node_by_output(
                    input_variable
                )

                if producer is not None:

                    # producer -> current node

                    if producer not in node.predecessors:

                        node.predecessors.append(
                            producer
                        )

                    if node not in producer.successors:

                        producer.successors.append(
                            node
                        )

    def topological_sort(self):

        # Calculate indegree of every node
        indegree = {}

        for node in self.nodes:

            indegree[node.node_id] = len(
                node.predecessors
            )

        # Nodes with no predecessors
        ready = []

        for node in self.nodes:

            if indegree[node.node_id] == 0:

                ready.append(node)
```

```
        order = []

        while ready:

            current = ready.pop(0)

            order.append(current)

            # Remove current node from the graph
            for successor in current.successors:

                indegree[successor.node_id] -= 1

                if indegree[successor.node_id] == 0:

                    ready.append(successor)

        # If not all nodes were processed,
        # the graph contains a cycle.
        if len(order) != len(self.nodes):

            raise ValueError(
                "DFG contains a cycle. "
                "The DFG must be acyclic."
            )

        return order

    def display(self):

        print("\n========== DFG ==========")

        print("\nPrimary Inputs:")
        print(self.inputs)

        print("\nOperations:")

        for node in self.nodes:

            print(
                f"{node.node_id}: "
                f"{node.operation} "
                f"{node.inputs} -> "
                f"{node.output}"
            )

        print("\nPrimary Outputs:")
        print(self.outputs)

        print("========================\n")
```

```
    def display_dependencies(self):

        print(
            "\n========== DEPENDENCY GRAPH =========="
        )

        for node in self.nodes:

            predecessors = [
                n.node_id
                for n in node.predecessors
            ]

            successors = [
                n.node_id
                for n in node.successors
            ]

            print(f"\nNode: {node.node_id}")

            print(
                f"  Operation    : "
                f"{node.operation}"
            )

            print(
                f"  Inputs       : "
                f"{node.inputs}"
            )

            print(
                f"  Output       : "
                f"{node.output}"
            )

            print(
                f"  Predecessors : "
                f"{predecessors}"
            )

            print(
                f"  Successors   : "
                f"{successors}"
            )

        print(
            "\n======================================\n"
        )
```

This file defines the internal Data Flow Graph data structure.
It contains:

- DFGNode — represents an individual operation.
- DFG — stores the complete graph.
- Dependency information through predecessors and successors.
- ASAP, ALAP and mobility information.
- Functional-unit binding information.
- Topological sorting.

Purpose: Acts as the central data structure used by almost every subsequent stage of the HLS tool.

3. main.py

```
from dfg import DFG, DFGNode

from scheduler import (
    asap_schedule,
    display_asap_schedule,
    alap_schedule,
    display_alap_schedule,
    get_latency
)

from resources import RESOURCE_LIMITS

from list_scheduler import (
    list_schedule,
    display_list_schedule
)

from resources import RESOURCE_LIMITS

from list_scheduler import (
    list_schedule,
    display_list_schedule
)

from fu_binding import (
    bind_functional_units,
    display_fu_binding
)

from register_binding import (
    find_variable_live_ranges,
    allocate_registers,
    display_register_binding
)

from datapath import (
    generate_datapath,
    display_datapath
)
```

```
from mux_control import (
    generate_mux_controls,
    display_mux_controls
)

from netlist import (
    generate_netlist,
    display_netlist
)

from verilog_generator import generate_verilog

def parse_dfg(filename):

    dfg = DFG()

    with open(filename, "r") as file:

        for line in file:

            line = line.strip()

            if not line:
                continue

            if line.startswith("#"):
                continue

            if line.startswith("INPUTS:"):

                data = line.split(":", 1)[1].strip()

                dfg.inputs = [
                    x.strip()
                    for x in data.split(",")
                ]

            elif line.startswith("NODE:"):

                data = line.split(":", 1)[1].strip()

                tokens = data.split()

                node_id = tokens[0]
                operation = tokens[1]

                inputs = tokens[2:-1]

                output = tokens[-1]

                node = DFGNode(
```

```
                    node_id,
                    operation,
                    inputs,
                    output
                )

                dfg.add_node(node)

            elif line.startswith("OUTPUTS:"):

                data = line.split(":", 1)[1].strip()

                dfg.outputs = [
                    x.strip()
                    for x in data.split(",")
                ]

    return dfg

def main():

    filename = "input/example.dfg"

    # --------------------------------
    # Step 1: Parse DFG
    # --------------------------------

    dfg = parse_dfg(filename)

    dfg.display()

    # --------------------------------
    # Step 2: Build dependencies
    # --------------------------------

    dfg.build_dependencies()

    dfg.display_dependencies()

    # --------------------------------
    # Topological ordering
    # --------------------------------

    print(
        "========== TOPOLOGICAL ORDER =========="
    )

    order = dfg.topological_sort()

    print(
        " -> ".join(
```

```
            node.node_id
            for node in order
        )
    )

    print(
        "========================================"
    )

    # --------------------------------
    # Step 3: ASAP scheduling
    # --------------------------------

    asap_schedule(dfg)

    display_asap_schedule(dfg)

    # --------------------------------
    # Calculate overall latency
    # --------------------------------

    latency = max(
        node.asap
        + get_latency(node.operation)
        - 1
        for node in dfg.nodes
    )

    print(
        f"\nOverall schedule latency: "
        f"{latency} cycles"
    )

    # --------------------------------
    # Step 4: ALAP scheduling
    # --------------------------------

    alap_schedule(
        dfg,
        latency
    )

    display_alap_schedule(dfg)

    # --------------------------------
    # Step 5: List scheduling
    # --------------------------------

    print(
        "\nResource constraints:"
    )
```

```
    for operation, count in RESOURCE_LIMITS.items():

        print(
            f"  {operation}: "
            f"{count}"
        )

    schedule, scheduled = list_schedule(
        dfg,
        RESOURCE_LIMITS
    )

    display_list_schedule(
        schedule,
        scheduled
    )

    # --------------------------------
    # Step 6: Functional Unit Binding
    # --------------------------------

    binding = bind_functional_units(
        schedule,
        RESOURCE_LIMITS
    )

    display_fu_binding(
        schedule,
        binding
    )

    # --------------------------------
    # Step 7: Register Binding
    # --------------------------------

    live_ranges = find_variable_live_ranges(
        dfg,
        scheduled
    )

    registers = allocate_registers(
        live_ranges
    )

    display_register_binding(
        live_ranges,
        registers
    )

    # --------------------------------
```

```
    # Step 8: Datapath Generation
    # --------------------------------

    datapath = generate_datapath(
        dfg,
        schedule,
        binding,
        registers
    )

    display_datapath(
        datapath
    )

    # --------------------------------
    # Step 9: MUX and Control Generation
    # --------------------------------

    controls = generate_mux_controls(
        dfg,
        schedule,
        binding,
        registers
    )

    display_mux_controls(
        controls
    )

    # --------------------------------
    # Step 10: Netlist Generation
    # --------------------------------

    netlist = generate_netlist(
        dfg,
        schedule,
        binding,
        registers,
        controls
    )

    display_netlist(
        netlist
    )

    # --------------------------------
    # Step 11: Verilog Generation
    # --------------------------------

    verilog_file = generate_verilog(
        dfg,
```

```
        schedule,
        binding,
        registers,
        controls
    )

    print(
        "\n========================================="
    )

    print(
        "VERILOG NETLIST GENERATED"
    )

    print(
        f"File: {verilog_file}"
    )

    print(
        "========================================="
    )

    # ===================================
    # Step 12: DFG Visualization
    # ===================================

    print(
        "\n========================================="
    )

    print(
        "GENERATING DFG VISUALIZATION"
    )

    print(
        "========================================="
    )

    try:

        import subprocess
        import sys

        subprocess.run(
            [
                sys.executable,
                "visualizer.py",
                filename
            ],
            check=True
        )
```

```
        print(
            "\nDFG VISUALIZATION GENERATED"
        )

        print(
            "Files:"
        )

        print(
            "  docs/dfg_auto.dot"
        )

        print(
            "  docs/dfg_auto.png"
        )

        print(
            "  docs/dfg_auto.svg"
        )

    except Exception as error:

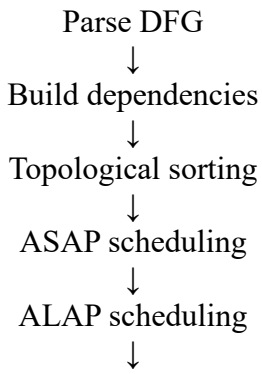
        print(
            "\nWARNING: DFG visualization failed."
        )

        print(
            f"Reason: {error}"
        )

# ====================================
# Program entry point
# ====================================

if __name__ == "__main__":
    main()
```

main.py is the **main driver program** that connects all stages of the project.



List scheduling
↓
FU binding
↓
Register binding
↓
MUX/control generation
↓
Datapath generation
↓
Netlist generation
↓
Verilog generation

Purpose: Orchestrates the complete HLS flow from the input DFG to generated RTL.

4. visualizer.py

```
import os
import re
import subprocess
import sys

# ============================================================
# DFG PARSER
# ============================================================

def parse_dfg(filename):
    """
    Parse the HLS DFG file.

    Expected format:

        INPUTS: a,b,c,d,e,f,g,h

        NODE: N1 ADD a b x1
        NODE: N2 ADD c d x2
        NODE: N3 MUL x1 x2 x3
        NODE: N4 ADD e f x4
        NODE: N5 ADD x3 x4 x5
        NODE: N6 MUL g h x6
        NODE: N7 ADD x5 x6 y

        OUTPUTS: y
    """

    operations = []
    primary_inputs = []
    declared_outputs = []

    with open(filename, "r") as file:
```

```
        for raw_line in file:

            line = raw_line.strip()

            # ------------------------------------------------
            # Ignore empty lines
            # ------------------------------------------------

            if not line:
                continue

            # ------------------------------------------------
            # Ignore comments
            # ------------------------------------------------

            if line.startswith("#"):
                continue

            # ------------------------------------------------
            # Parse INPUTS
            # ------------------------------------------------

            if line.startswith("INPUTS:"):

                values = line.split(":", 1)[1]

                primary_inputs = [
                    x.strip()
                    for x in values.split(",")
                    if x.strip()
                ]

                continue

            # ------------------------------------------------
            # Parse OUTPUTS
            # ------------------------------------------------

            if line.startswith("OUTPUTS:"):

                values = line.split(":", 1)[1]

                declared_outputs = [
                    x.strip()
                    for x in values.split(",")
                    if x.strip()
                ]

                continue
```

```
            # ------------------------------------------------
            # Parse NODE
            # ------------------------------------------------

            if line.startswith("NODE:"):

                parts = line.split()

                # Expected:
                #
                # NODE: N1 ADD a b x1
                #
                # parts:
                # 0      1  2   3 4 5
                # NODE:  N1 ADD a b x1

                if len(parts) != 6:

                    print(
                        f"WARNING: Invalid NODE line:\n"
                        f"  {line}"
                    )

                    continue

                node = parts[1]
                operation = parts[2].upper()
                input_a = parts[3]
                input_b = parts[4]
                output = parts[5]

                operations.append({
                    "node": node,
                    "op": operation,
                    "inputs": [input_a, input_b],
                    "output": output
                })

    return operations, primary_inputs, declared_outputs

# ============================================================
# GRAPHVIZ NODE STYLE
# ============================================================

def get_operation_style(operation):

    operation = operation.upper()

    if operation == "ADD":

        return (
```

```
            "ADD",
            "lightblue"
        )

    if operation == "SUB":

        return (
            "SUB",
            "lightblue"
        )

    if operation == "MUL":

        return (
            "MUL",
            "lightgreen"
        )

    if operation == "DIV":

        return (
            "DIV",
            "lightyellow"
        )

    return (
        operation,
        "lightgray"
    )

# ============================================================
# GENERATE DOT
# ============================================================

def generate_dot(
    operations,
    primary_inputs,
    declared_outputs,
    dot_file
):

    lines = []

    # ------------------------------------------------------------
    # Graph header
    # ------------------------------------------------------------

    lines.append("digraph DFG {")
    lines.append("")
    lines.append("    rankdir=LR;")
    lines.append("    splines=ortho;")
```

```
    lines.append("    nodesep=0.7;")
    lines.append("    ranksep=1.0;")
    lines.append("")

    lines.append("    graph [")
    lines.append('        fontname="Arial",')
    lines.append(
        '        label="Data Flow Graph (DFG) — Automatic Visualization",'
    )
    lines.append("        labelloc=t,")
    lines.append("        fontsize=20,")
    lines.append("        pad=0.3")
    lines.append("    ];")
    lines.append("")

    lines.append("    node [")
    lines.append('        fontname="Arial",')
    lines.append("        fontsize=10")
    lines.append("    ];")
    lines.append("")

    lines.append("    edge [")
    lines.append('        fontname="Arial",')
    lines.append("        fontsize=9,")
    lines.append("        arrowsize=0.7")
    lines.append("    ];")
    lines.append("")

    # ------------------------------------------------------------
    # Primary input nodes
    # ------------------------------------------------------------

    for signal in primary_inputs:

        safe_name = re.sub(
            r"[^A-Za-z0-9_]",
            "_",
            signal
        )

        lines.append(
            f'    IN_{safe_name} '
            f'[shape=circle, '
            f'label="{signal}", '
            f'fillcolor="white", '
            f'style="filled"];'
        )

    lines.append("")

    # ------------------------------------------------------------
```

```
    # Operation nodes
    # ---------------------------------------------------

    for item in operations:

        node = item["node"]
        operation = item["op"]
        output = item["output"]

        operation_label, fill_color = get_operation_style(
            operation
        )

        label = (
            f"{node}\\n"
            f"{operation_label}\\n"
            f"{output}"
        )

        lines.append(
            f'    {node} ['
            f'shape=box, '
            f'style="rounded,filled", '
            f'label="{label}", '
            f'fillcolor="{fill_color}", '
            f'penwidth=1.5'
            f'];'
        )

    lines.append("")

    # ---------------------------------------------------------
    # Determine outputs
    # ---------------------------------------------------------

    produced_values = {
        item["output"]
        for item in operations
    }

    consumed_values = {
        signal
        for item in operations
        for signal in item["inputs"]
    }

    # Use explicitly declared outputs when available.
    if declared_outputs:

        final_outputs = declared_outputs
```

```
    else:

        final_outputs = sorted(
            produced_values - consumed_values
        )

    # ------------------------------------------------------
    # Output nodes
    # ------------------------------------------------------

    for signal in final_outputs:

        safe_name = re.sub(
            r"[^A-Za-z0-9_]",
            "_",
            signal
        )

        lines.append(
            f'    OUT_{safe_name} '
            f'[shape=doublecircle, '
            f'label="{signal}", '
            f'fillcolor="lightcyan", '
            f'style="filled"];'
        )

    lines.append("")

    # ------------------------------------------------------
    # Map produced signal → generating node
    # ------------------------------------------------------

    producer = {
        item["output"]: item["node"]
        for item in operations
    }

    # ------------------------------------------------------
    # Create data-flow edges
    # ------------------------------------------------------

    for item in operations:

        destination_node = item["node"]

        for index, signal in enumerate(item["inputs"]):

            # ----------------------------------------------
            # Signal comes from another operation
            # ----------------------------------------------
```

```
            if signal in producer:

                source_node = producer[signal]

                lines.append(
                    f'    {source_node} -> {destination_node} '
                    f'[xlabel="{signal}"];'
                )

            # ------------------------------------------------
            # Signal is a primary input
            # ------------------------------------------------

            else:

                safe_name = re.sub(
                    r"[^A-Za-z0-9_]",
                    "_",
                    signal
                )

                lines.append(
                    f'    IN_{safe_name} -> {destination_node};'
                )

    lines.append("")

    # ------------------------------------------------------------
    # Connect final outputs
    # ------------------------------------------------------------

    for signal in final_outputs:

        if signal not in producer:
            continue

        source_node = producer[signal]

        safe_name = re.sub(
            r"[^A-Za-z0-9_]",
            "_",
            signal
        )

        lines.append(
            f'    {source_node} -> OUT_{safe_name};'
        )

    lines.append("")

    # ------------------------------------------------------------
```

```
    # Close graph
    # ------------------------------------------------------

    lines.append("}")

    with open(dot_file, "w") as file:

        file.write(
            "\n".join(lines)
        )

# ============================================================
# GENERATE PNG / SVG
# ============================================================

def generate_images(dot_file, output_directory):

    png_file = os.path.join(
        output_directory,
        "dfg_auto.png"
    )

    svg_file = os.path.join(
        output_directory,
        "dfg_auto.svg"
    )

    # ------------------------------------------------------
    # PNG
    # ------------------------------------------------------

    subprocess.run(
        [
            "dot",
            "-Tpng",
            dot_file,
            "-o",
            png_file
        ],
        check=True
    )

    # ------------------------------------------------------
    # SVG
    # ------------------------------------------------------

    subprocess.run(
        [
            "dot",
            "-Tsvg",
            dot_file,
```

```
            "-o",
            svg_file
        ],
        check=True
    )

    return png_file, svg_file

# ============================================================
# MAIN
# ============================================================

def main():

    # ------------------------------------------------------
    # Check command-line argument
    # ------------------------------------------------------

    if len(sys.argv) != 2:

        print()
        print(
            "Usage:"
        )
        print(
            "  python3 visualizer.py input/example.dfg"
        )
        print()

        sys.exit(1)

    dfg_file = sys.argv[1]

    # ------------------------------------------------------
    # Check DFG file
    # ------------------------------------------------------

    if not os.path.isfile(dfg_file):

        print()
        print(
            f"ERROR: DFG file not found:"
        )
        print(
            f"  {dfg_file}"
        )
        print()

        sys.exit(1)

    # ------------------------------------------------------
```

```
    # Create docs directory
    # ------------------------------------------------------

    docs_directory = "docs"

    os.makedirs(
        docs_directory,
        exist_ok=True
    )

    # ------------------------------------------------------
    # Parse DFG
    # ------------------------------------------------------

    operations, primary_inputs, declared_outputs = parse_dfg(
        dfg_file
    )

    # ------------------------------------------------------
    # Validate
    # ------------------------------------------------------

    if not operations:

        print()
        print(
            "ERROR: No DFG operations found."
        )
        print()
        print(
            "Expected format:"
        )
        print(
            "  NODE: N1 ADD a b x1"
        )
        print(
            "  NODE: N2 MUL x1 x2 x3"
        )
        print()

        sys.exit(1)

    # ------------------------------------------------------
    # Generate DOT
    # ------------------------------------------------------

    dot_file = os.path.join(
        docs_directory,
        "dfg_auto.dot"
    )
```

```
    generate_dot(
        operations,
        primary_inputs,
        declared_outputs,
        dot_file
    )

    # ------------------------------------------------------
    # Generate images
    # ------------------------------------------------------

    png_file, svg_file = generate_images(
        dot_file,
        docs_directory
    )

    # ------------------------------------------------------
    # Print summary
    # ------------------------------------------------------

    print()
    print("================================================")
    print("           DFG VISUALIZATION")
    print("================================================")
    print()
    print(f"Input file : {dfg_file}")
    print()
    print(f"Operations : {len(operations)}")
    print(
        f"Inputs     : {', '.join(primary_inputs)}"
    )
    print(
        f"Outputs    : {', '.join(declared_outputs)}"
    )
    print()
    print(f"DOT file   : {dot_file}")
    print(f"PNG file   : {png_file}")
    print(f"SVG file   : {svg_file}")
    print()
    print("================================================")

# ============================================================
# PROGRAM ENTRY POINT
# ============================================================

if __name__ == "__main__":
    main()
```

This file parses the DFG and generates a **Graphviz representation** of the computation graph.
It produces visual outputs such as:

docs/dfg_auto.png
docs/dfg_auto.svg

Purpose**:** Provides a graphical representation of the computation and dependencies, making the DFG easier to analyze.

5. scheduler.py

```
from collections import defaultdict

OPERATION_LATENCY = {
    "ADD": 1,
    "SUB": 1,
    "MUL": 1,
    "DIV": 3
}

def get_latency(operation):

    if operation not in OPERATION_LATENCY:

        raise ValueError(
            f"Unknown operation: {operation}"
        )

    return OPERATION_LATENCY[operation]

def asap_schedule(dfg):

    # Obtain dependency-respecting order
    order = dfg.topological_sort()

    for node in order:

        # No predecessors
        if not node.predecessors:

            node.asap = 1

        else:

            earliest_start = 1

            for predecessor in node.predecessors:

                predecessor_latency = get_latency(
                    predecessor.operation
                )

                predecessor_finish = (
                    predecessor.asap
                    + predecessor_latency
```

```
                )

                earliest_start = max(
                    earliest_start,
                    predecessor_finish
                )

            node.asap = earliest_start

    return order

def display_asap_schedule(dfg):

    print("\n========== ASAP SCHEDULE ==========")

    schedule = defaultdict(list)

    for node in dfg.nodes:

        schedule[node.asap].append(node)

    for cycle in sorted(schedule):

        operations = schedule[cycle]

        names = [
            node.node_id
            for node in operations
        ]

        print(
            f"Cycle {cycle}: "
            + ", ".join(names)
        )

    print("====================================\n")

def alap_schedule(dfg, latency):

    # Get reverse topological order
    order = dfg.topological_sort()

    reverse_order = list(reversed(order))

    for node in reverse_order:

        # If node has no successors,
        # it must finish by the final latency.
        if not node.successors:

            node.alap = (
```

```
                latency
                - get_latency(node.operation)
                + 1
            )

        else:

            latest_start = float("inf")

            for successor in node.successors:

                successor_alap = successor.alap

                node_latest_start = (
                    successor_alap
                    - get_latency(node.operation)
                )

                latest_start = min(
                    latest_start,
                    node_latest_start
                )

            node.alap = latest_start

    # Calculate mobility
    for node in dfg.nodes:

        node.mobility = (
            node.alap - node.asap
        )

    return order

def display_alap_schedule(dfg):

    print("\n========== ASAP / ALAP / MOBILITY ==========")

    print(
        f"{'Node':<8}"
        f"{'Operation':<12}"
        f"{'ASAP':<8}"
        f"{'ALAP':<8}"
        f"{'Mobility':<10}"
    )

    print("-" * 46)

    for node in dfg.nodes:

        print(
```

```
            f"{node.node_id:<8}"
            f"{node.operation:<12}"
            f"{node.asap:<8}"
            f"{node.alap:<8}"
            f"{node.mobility:<10}"
        )

    print("=" * 46)
```

This file implements the basic scheduling analysis.
It performs:

1. Operation latency determination
2. ASAP scheduling
3. ALAP scheduling
4. Mobility calculation

The scheduling information is used to determine when each operation can execute.
Purpose**:** Establishes the temporal constraints of the DFG before resource-constrained scheduling.

6. resources.py

```
RESOURCE_LIMITS = {
    "MUL": 1,
    "ADD": 1,
    "SUB": 1,
    "DIV": 1
}
```

This file specifies the number of functional units available to the scheduler.
For example:

ADD → 1
MUL → 1
SUB → 1
DIV → 1

Purpose**:** Defines the hardware resource limitations that influence resource-constrained scheduling.

7. list_scheduler.py

```
from collections import defaultdict

from scheduler import get_latency

def is_ready(node, scheduled):

    for predecessor in node.predecessors:

        if predecessor.node_id not in scheduled:
            return False

    return True

def list_schedule(
    dfg,
```

```
    resource_limits
):

    # Nodes that have already been scheduled
    scheduled = {}

    # Final schedule
    schedule = defaultdict(list)

    # Current clock cycle
    cycle = 1

    while len(scheduled) < len(dfg.nodes):

        # --------------------------------
        # Find ready operations
        # --------------------------------

        ready = []

        for node in dfg.nodes:

            if node.node_id in scheduled:
                continue

            if is_ready(node, scheduled):

                ready.append(node)

        # --------------------------------
        # Sort ready operations
        #
        # Priority:
        # 1. Mobility
        # 2. ASAP time
        # 3. Node ID
        # --------------------------------

        ready.sort(
            key=lambda node: (
                node.mobility,
                node.asap,
                node.node_id
            )
        )

        # --------------------------------
        # Track resources used
        # --------------------------------

        resources_used = defaultdict(int)
```

```
        # --------------------------------
        # Schedule ready operations
        # --------------------------------

        for node in ready:

            operation = node.operation

            limit = resource_limits.get(
                operation,
                0
            )

            if resources_used[operation] >= limit:
                continue

            # Schedule node
            schedule[cycle].append(node)

            scheduled[node.node_id] = cycle

            resources_used[operation] += 1

        # --------------------------------
        # Safety check
        # --------------------------------

        if not schedule[cycle]:

            raise RuntimeError(
                f"No operation could be scheduled "
                f"in cycle {cycle}. "
                f"Check resource constraints."
            )

        cycle += 1

    return schedule, scheduled

def display_list_schedule(
    schedule,
    scheduled
):

    print(
        "\n========== LIST SCHEDULE =========="
    )

    for cycle in sorted(schedule):
```

```
        nodes = schedule[cycle]

        names = [
            node.node_id
            for node in nodes
        ]

        print(
            f"Cycle {cycle}: "
            + ", ".join(names)
        )

    print(
        "====================================\n"
    )

    print(
        "========== OPERATION TIMES =========="
    )

    for node_id in sorted(scheduled):

        print(
            f"{node_id}: "
            f"Cycle {scheduled[node_id]}"
        )

    print(
        "====================================\n"
    )
```

This file performs list scheduling while considering the available hardware resources.
It determines:

- Which operations are ready
- Which operations can execute in each cycle
- Resource availability
- Operation priority
- Final execution cycle

Purpose: Converts the dependency-constrained DFG into a practical cycle-by-cycle schedule under hardware resource limitations.

8. fu_binding.py

```
from collections import defaultdict

def bind_functional_units(schedule, resource_limits):

    # Available physical functional units
    available_units = {}
```

```
    for operation, count in resource_limits.items():

        available_units[operation] = [
            f"{operation}{i + 1}"
            for i in range(count)
        ]

    # Store binding
    binding = {}

    # Process schedule cycle by cycle
    for cycle in sorted(schedule):

        # Track which units are already used
        used_units = defaultdict(set)

        for node in schedule[cycle]:

            operation = node.operation

            units = available_units.get(
                operation,
                []
            )

            if not units:

                raise RuntimeError(
                    f"No functional unit available "
                    f"for operation {operation}"
                )

            # Find a free unit
            selected_unit = None

            for unit in units:

                if unit not in used_units[operation]:

                    selected_unit = unit
                    break

            if selected_unit is None:

                raise RuntimeError(
                    f"Resource conflict in cycle "
                    f"{cycle} for {operation}"
                )

            binding[node.node_id] = selected_unit
```

```
            used_units[operation].add(
                selected_unit
            )

            # Store binding in DFG node
            node.functional_unit = selected_unit

    return binding

def display_fu_binding(
    schedule,
    binding
):

    print(
        "\n========== FUNCTIONAL UNIT BINDING =========="
    )

    print(
        f"{'Cycle':<8}"
        f"{'Node':<8}"
        f"{'Operation':<12}"
        f"{'Functional Unit':<18}"
    )

    print("-" * 46)

    for cycle in sorted(schedule):

        for node in schedule[cycle]:

            print(
                f"{cycle:<8}"
                f"{node.node_id:<8}"
                f"{node.operation:<12}"
                f"{binding[node.node_id]:<18}"
            )

    print("=" * 46)

    print(
        "\nPhysical Functional Units:"
    )

    units = sorted(
        set(binding.values())
    )

    for unit in units:

        operations = [
```

```
            node_id
            for node_id, fu in binding.items()
            if fu == unit
        ]

        print(
            f"  {unit}: "
            + ", ".join(operations)
        )

    print(
        "============================================\n"
    )
```

After scheduling, this file assigns each operation to a specific physical functional unit.
For example:

ADD operations → ADD1
MUL operations → MUL1

Multiple operations can share the same functional unit at different clock cycles.
Purpose**:** Performs hardware resource sharing through functional-unit binding.

9. register_binding.py

```
def find_variable_live_ranges(
    dfg,
    scheduled
):

    live_ranges = {}

    for node in dfg.nodes:

        variable = node.output

        # Ignore final primary output
        # for register allocation
        if variable in dfg.outputs:
            continue

        production_cycle = scheduled[
            node.node_id
        ]

        # Find all consumers
        consumer_cycles = []

        for successor in node.successors:

            consumer_cycles.append(
                scheduled[successor.node_id]
```

```
            )

        if consumer_cycles:

            last_use = max(
                consumer_cycles
            )

        else:

            # Value is never consumed
            last_use = production_cycle + 1

        live_ranges[variable] = (
            production_cycle,
            last_use
        )

    return live_ranges

def live_ranges_overlap(
    range1,
    range2
):

    start1, end1 = range1
    start2, end2 = range2

    return (
        start1 < end2
        and start2 < end1
    )

def allocate_registers(
    live_ranges
):

    registers = {}

    # Process variables by
    # production time
    variables = sorted(
        live_ranges,
        key=lambda variable: (
            live_ranges[variable][0],
            variable
        )
    )

    for variable in variables:
```

```
        assigned = False

        for register in sorted(
            set(registers.values())
        ):

            conflict = False

            for other_variable, other_register in registers.items():

                if other_register != register:
                    continue

                if live_ranges_overlap(
                    live_ranges[variable],
                    live_ranges[other_variable]
                ):

                    conflict = True
                    break

            if not conflict:

                registers[variable] = register
                assigned = True
                break

        if not assigned:

            register_number = (
                len(set(registers.values()))
                + 1
            )

            register = f"R{register_number}"

            registers[variable] = register

    return registers

def display_register_binding(
    live_ranges,
    registers
):

    print(
        "\n========== REGISTER BINDING =========="
    )

    print(
        f"{'Variable':<12}"
```

```
        f"{'Live Range':<15}"
        f"{'Register':<10}"
    )

    print("-" * 37)

    for variable in sorted(
        live_ranges,
        key=lambda x: live_ranges[x][0]
    ):

        start, end = live_ranges[
            variable
        ]

        print(
            f"{variable:<12}"
            f"[{start}, {end})"
            f"{'':<8}"
            f"{registers[variable]:<10}"
        )

    print("=" * 37)

    print(
        "\nPhysical Registers:"
    )

    for register in sorted(
        set(registers.values())
    ):

        variables = [
            variable
            for variable, reg
            in registers.items()
            if reg == register
        ]

        print(
            f"  {register}: "
            + ", ".join(variables)
        )

    print(
        "=======================================\n"
    )
```

This file determines how intermediate variables are stored in physical registers.
It performs:

- Variable live-range analysis
- Register allocation
- Register reuse

For example:

x1 → R1
x2 → R2
x3 → R3

Purpose**:** Maps logical intermediate variables to the minimum practical set of physical registers.

10. mux_controller.py

```
def variable_to_source(variable, register_binding):

    # Intermediate variable stored in a register
    if variable in register_binding:
        return register_binding[variable]

    # Primary input
    return variable

def generate_mux_controls(
    dfg,
    schedule,
    fu_binding,
    register_binding
):

    controls = []

    for cycle in sorted(schedule):

        for node in schedule[cycle]:

            fu = fu_binding[node.node_id]

            input_a = variable_to_source(
                node.inputs[0],
                register_binding
            )

            input_b = variable_to_source(
                node.inputs[1],
                register_binding
            )

            control = {
                "cycle": cycle,
                "node": node.node_id,
                "functional_unit": fu,
                "input_a": input_a,
                "input_b": input_b,
                "output": node.output
```

```
            }

            controls.append(control)

    return controls

def display_mux_controls(controls):

    print(
        "\n========== MUX / CONTROL SIGNALS =========="
    )

    print(
        f"{'Cycle':<8}"
        f"{'Node':<8}"
        f"{'FU':<8}"
        f"{'Input A':<12}"
        f"{'Input B':<12}"
        f"{'Output':<10}"
    )

    print("-" * 58)

    for control in controls:

        print(
            f"{control['cycle']:<8}"
            f"{control['node']:<8}"
            f"{control['functional_unit']:<8}"
            f"{control['input_a']:<12}"
            f"{control['input_b']:<12}"
            f"{control['output']:<10}"
        )

    print("=" * 58)
```

This file determines which values must be selected at the inputs of the functional units during each clock cycle. For example:

Cycle 1:
ADD1 ← a,b

Cycle 2:
ADD1 ← c,d

Cycle 3:
MUL1 ← R1,R2

Purpose**:** Generates the control information required to connect registers/inputs to functional units through multiplexers.

11. datapath.py

```
def generate_datapath(
    dfg,
    schedule,
    fu_binding,
    register_binding
):

    datapath = {
        "inputs": list(dfg.inputs),
        "outputs": list(dfg.outputs),
        "functional_units": sorted(
            set(fu_binding.values())
        ),
        "registers": sorted(
            set(register_binding.values())
        ),
        "operations": [],
        "connections": []
    }

    # --------------------------------
    # Generate operation information
    # --------------------------------

    for cycle in sorted(schedule):

        for node in schedule[cycle]:

            operation = {
                "cycle": cycle,
                "node": node.node_id,
                "operation": node.operation,
                "functional_unit":
                    fu_binding[node.node_id],
                "inputs": list(node.inputs),
                "output": node.output
            }

            # Register receiving the result
            if node.output in register_binding:

                operation["destination_register"] = (
                    register_binding[node.output]
                )

            else:

                operation["destination_register"] = (
                    "OUTPUT"
                )
```

```
            datapath["operations"].append(
                operation
            )

    return datapath

def display_datapath(datapath):

    print(
        "\n========== DATAPATH =========="
    )

    print("\nInputs:")

    print(
        "  "
        + ", ".join(datapath["inputs"])
    )

    print("\nFunctional Units:")

    for unit in datapath["functional_units"]:

        print(
            f"  {unit}"
        )

    print("\nRegisters:")

    for register in datapath["registers"]:

        print(
            f"  {register}"
        )

    print("\nOperations:")

    print(
        f"{'Cycle':<8}"
        f"{'Node':<8}"
        f"{'Operation':<10}"
        f"{'FU':<8}"
        f"{'Inputs':<20}"
        f"{'Destination':<12}"
    )

    print("-" * 66)

    for operation in datapath["operations"]:
```

```
        inputs = ", ".join(
            operation["inputs"]
        )

        print(
            f"{operation['cycle']:<8}"
            f"{operation['node']:<8}"
            f"{operation['operation']:<10}"
            f"{operation['functional_unit']:<8}"
            f"{inputs:<20}"
            f"{operation['destination_register']:<12}"
        )

    print(
        "=" * 66
    )
```

This file combines the results of:

- Scheduling
- Functional-unit binding
- Register binding
- MUX/control generation

to construct the bound hardware datapath.
The resulting architecture contains:

Registers
+
Functional Units
+
MUXes
+
Connections

Purpose: Converts the abstract scheduling/binding decisions into a hardware datapath representation.

12. netlist.py

```
def generate_netlist(
    dfg,
    schedule,
    fu_binding,
    register_binding,
    controls
):

    netlist = {
        "inputs": list(dfg.inputs),
        "outputs": list(dfg.outputs),
        "registers": sorted(
            set(register_binding.values())
        ),
        "functional_units": sorted(
            set(fu_binding.values())
```

```
        ),
        "muxes": [],
        "connections": [],
        "operations": []
    }

    # --------------------------------
    # Generate MUXes
    # --------------------------------

    fu_inputs = {}

    for control in controls:

        fu = control["functional_unit"]

        if fu not in fu_inputs:
            fu_inputs[fu] = {
                "A": set(),
                "B": set()
            }

        fu_inputs[fu]["A"].add(
            control["input_a"]
        )

        fu_inputs[fu]["B"].add(
            control["input_b"]
        )

    for fu in sorted(fu_inputs):

        netlist["muxes"].append({
            "name": f"{fu}_A_MUX",
            "functional_unit": fu,
            "port": "A",
            "inputs": sorted(
                fu_inputs[fu]["A"]
            )
        })

        netlist["muxes"].append({
            "name": f"{fu}_B_MUX",
            "functional_unit": fu,
            "port": "B",
            "inputs": sorted(
                fu_inputs[fu]["B"]
            )
        })

    # --------------------------------
```

```
    # Generate operation connections
    # --------------------------------

    for control in controls:

        operation = {
            "cycle": control["cycle"],
            "node": control["node"],
            "functional_unit":
                control["functional_unit"],
            "input_a": control["input_a"],
            "input_b": control["input_b"],
            "output": control["output"]
        }

        # Determine destination
        output = control["output"]

        if output in register_binding:

            operation["destination"] = (
                register_binding[output]
            )

        else:

            operation["destination"] = (
                "OUTPUT"
            )

        netlist["operations"].append(
            operation
        )

    return netlist

def display_netlist(netlist):

    print(
        "\n========== GENERATED NETLIST =========="
    )

    print("\nPRIMARY INPUTS:")

    for signal in netlist["inputs"]:

        print(
            f"  INPUT {signal}"
        )

    print("\nREGISTERS:")
```

```
    for register in netlist["registers"]:

        print(
            f"  REGISTER {register}"
        )

    print("\nFUNCTIONAL UNITS:")

    for unit in netlist["functional_units"]:

        print(
            f"  {unit}"
        )

    print("\nMULTIPLEXERS:")

    for mux in netlist["muxes"]:

        inputs = ", ".join(
            mux["inputs"]
        )

        print(
            f"  {mux['name']}: "
            f"{inputs}"
        )

    print("\nOPERATIONS / CONNECTIONS:")

    print(
        f"{'Cycle':<7}"
        f"{'Node':<7}"
        f"{'FU':<7}"
        f"{'A':<8}"
        f"{'B':<8}"
        f"{'DEST':<10}"
    )

    print("-" * 57)

    for operation in netlist["operations"]:

        print(
            f"{operation['cycle']:<7}"
            f"{operation['node']:<7}"
            f"{operation['functional_unit']:<7}"
            f"{operation['input_a']:<8}"
            f"{operation['input_b']:<8}"
            f"{operation['destination']:<10}"
        )
```

```
    print(
        "\nOUTPUTS:"
    )

    for output in netlist["outputs"]:

        print(
            f"  OUTPUT {output}"
        )

    print(
        "========================================\n"
    )
```

This file creates a structured representation of the generated hardware.
The netlist describes:

1. Primary inputs
2. Registers
3. Functional units
4. Multiplexers
5. Operations
6. Connections
7. Outputs

Purpose**:** Provides an intermediate hardware representation between datapath construction and RTL generation.

13. verilog_generator.py

```
def generate_verilog(
    dfg,
    schedule,
    fu_binding,
    register_binding,
    controls,
    filename="output/datapath.v"
):

    lines = []

    # ================================================
    # Module declaration
    # ================================================

    lines.append("`timescale 1ns/1ps")
    lines.append("")
    lines.append("module generated_datapath #(")
    lines.append("    parameter WIDTH = 16")
    lines.append(")(")

    # Primary inputs
    input_ports = []
```

```
    for signal in dfg.inputs:
        input_ports.append(
            f"    input  logic [WIDTH-1:0] {signal}"
        )

    input_ports.append(
        "    input  logic clk"
    )

    input_ports.append(
        "    input  logic reset"
    )

    input_ports.append(
        "    output logic [WIDTH-1:0] y"
    )

    lines.append(",\n".join(input_ports))
    lines.append(");")
    lines.append("")

    # ================================================
    # Registers
    # ================================================

    lines.append("    // ============================================")
    lines.append("    // Physical Registers")
    lines.append("    // ============================================")

    for register in sorted(
        set(register_binding.values())
    ):

        lines.append(
            f"    logic [WIDTH-1:0] {register};"
        )

    lines.append("")

    # ================================================
    # Cycle counter
    # ================================================

    lines.append("    // ============================================")
    lines.append("    // Control / Cycle Counter")
    lines.append("    // ============================================")

    lines.append(
        "    logic [2:0] cycle;"
    )
```

```
    lines.append("")

    lines.append(
        "    always_ff @(posedge clk) begin"
    )

    lines.append(
        "        if (reset)"
    )

    lines.append(
        "            cycle <= 3'd1;"
    )

    lines.append(
        "        else if (cycle < 3'd5)"
    )

    lines.append(
        "            cycle <= cycle + 3'd1;"
    )

    lines.append(
        "        else"
    )

    lines.append(
        "            cycle <= 3'd5;"
    )

    lines.append(
        "    end"
    )

    lines.append("")

    # ================================================
    # MUX outputs
    # ================================================

    lines.append("    // ============================================")
    lines.append("    // MUX Outputs")
    lines.append("    // ============================================")

    lines.append(
        "    logic [WIDTH-1:0] add_a;"
    )

    lines.append(
        "    logic [WIDTH-1:0] add_b;"
```

```
    )

    lines.append(
        "    logic [WIDTH-1:0] mul_a;"
    )

    lines.append(
        "    logic [WIDTH-1:0] mul_b;"
    )

    lines.append("")

    # ================================================
    # Functional Unit Outputs
    # ================================================

    lines.append(
        "    logic [WIDTH-1:0] add_result;"
    )

    lines.append(
        "    logic [WIDTH-1:0] mul_result;"
    )

    lines.append("")

    # ================================================
    # ADD input MUX A
    # ================================================

    lines.append("    // ADD1 input A MUX")

    lines.append(
        "    always_comb begin"
    )

    lines.append(
        "        case (cycle)"
    )

    lines.append(
        f"            3'd1: add_a = a;"
    )

    lines.append(
        f"            3'd2: add_a = c;"
    )

    lines.append(
        f"            3'd3: add_a = e;"
    )
```

```
    lines.append(
        f"            3'd4: add_a = R1;"
    )

    lines.append(
        f"            3'd5: add_a = R1;"
    )

    lines.append(
        "            default: add_a = '0;"
    )

    lines.append(
        "        endcase"
    )

    lines.append(
        "    end"
    )

    lines.append("")

    # ================================================
    # ADD input MUX B
    # ================================================

    lines.append("    // ADD1 input B MUX")

    lines.append(
        "    always_comb begin"
    )

    lines.append(
        "        case (cycle)"
    )

    lines.append(
        "            3'd1: add_b = b;"
    )

    lines.append(
        "            3'd2: add_b = d;"
    )

    lines.append(
        "            3'd3: add_b = f;"
    )

    lines.append(
        "            3'd4: add_b = R3;"
```

```
    )

    lines.append(
        "            3'd5: add_b = R2;"
    )

    lines.append(
        "            default: add_b = '0;"
    )

    lines.append(
        "        endcase"
    )

    lines.append(
        "    end"
    )

    lines.append("")

    # ================================================
    # MUL input MUX A
    # ================================================

    lines.append("    // MUL1 input A MUX")

    lines.append(
        "    always_comb begin"
    )

    lines.append(
        "        case (cycle)"
    )

    lines.append(
        "            3'd1: mul_a = g;"
    )

    lines.append(
        "            3'd3: mul_a = R1;"
    )

    lines.append(
        "            default: mul_a = '0;"
    )

    lines.append(
        "        endcase"
    )

    lines.append(
```

```
        "    end"
    )

    lines.append("")

    # ================================================
    # MUL input MUX B
    # ================================================

    lines.append("    // MUL1 input B MUX")

    lines.append(
        "    always_comb begin"
    )

    lines.append(
        "        case (cycle)"
    )

    lines.append(
        "            3'd1: mul_b = h;"
    )

    lines.append(
        "            3'd3: mul_b = R3;"
    )

    lines.append(
        "            default: mul_b = '0;"
    )

    lines.append(
        "        endcase"
    )

    lines.append(
        "    end"
    )

    lines.append("")

    # ================================================
    # Functional Units
    # ================================================

    lines.append(
        "    // ============================================="
    )

    lines.append(
        "    // Functional Units"
```

```
    )

    lines.append(
        "    // ============================================="
    )

    lines.append(
        "    assign add_result = add_a + add_b;"
    )

    lines.append(
        "    assign mul_result = mul_a * mul_b;"
    )

    lines.append("")

    # ================================================
    # Register write logic
    # ================================================

    lines.append(
        "    // ============================================="
    )

    lines.append(
        "    // Register Write / Output Logic"
    )

    lines.append(
        "    // ============================================="
    )

    lines.append(
        "    always_ff @(posedge clk) begin"
    )

    lines.append(
        "        if (reset) begin"
    )

    for register in sorted(
        set(register_binding.values())
    ):

        lines.append(
            f"            {register} <= '0;"
        )

    lines.append(
        "            y <= '0;"
    )
```

```
    lines.append(
        "        end"
    )

    lines.append(
        "        else begin"
    )

    # Cycle 1
    lines.append(
        "            if (cycle == 3'd1) begin"
    )

    lines.append(
        "                R1 <= add_result;"
    )

    lines.append(
        "                R2 <= mul_result;"
    )

    lines.append(
        "            end"
    )

    # Cycle 2
    lines.append(
        "            else if (cycle == 3'd2) begin"
    )

    lines.append(
        "                R3 <= add_result;"
    )

    lines.append(
        "            end"
    )

    # Cycle 3
    lines.append(
        "            else if (cycle == 3'd3) begin"
    )

    lines.append(
        "                R1 <= mul_result;"
    )

    lines.append(
        "                R3 <= add_result;"
    )
```

```
    lines.append(
        "            end"
    )

    # Cycle 4
    lines.append(
        "            else if (cycle == 3'd4) begin"
    )

    lines.append(
        "                R1 <= add_result;"
    )

    lines.append(
        "            end"
    )

    # Cycle 5
    lines.append(
        "            else if (cycle == 3'd5) begin"
    )

    lines.append(
        "                y <= add_result;"
    )

    lines.append(
        "            end"
    )

    lines.append(
        "        end"
    )

    lines.append(
        "    end"
    )

    lines.append("")

    lines.append(
        "endmodule"
    )

    # ================================================
    # Write file
    # ================================================

    with open(filename, "w") as file:
```

```
        file.write(
            "\n".join(lines)
        )

    return filename
```

This file converts the HLS results into synthesizable SystemVerilog/Verilog RTL.
It generates:

output/datapath.v

The generated RTL contains:

- Module declaration
- Input/output ports
- Registers
- Cycle counter
- MUX logic
- Functional units
- Sequential logic
- Output logic

Purpose**:** Automatically translates the synthesized datapath into RTL suitable for simulation and subsequent synthesis.

14. tb/tb_datapath.sv

```
`timescale 1ns/1ps

module tb_datapath;

    parameter WIDTH = 16;

    logic clk;
    logic reset;

    logic [WIDTH-1:0] a;
    logic [WIDTH-1:0] b;
    logic [WIDTH-1:0] c;
    logic [WIDTH-1:0] d;
    logic [WIDTH-1:0] e;
    logic [WIDTH-1:0] f;
    logic [WIDTH-1:0] g;
    logic [WIDTH-1:0] h;

    logic [WIDTH-1:0] y;


    // ========================================
    // DUT
    // ========================================

    generated_datapath #(
        .WIDTH(WIDTH)
    ) dut (
```

```
        .clk(clk),
        .reset(reset),

        .a(a),
        .b(b),
        .c(c),
        .d(d),
        .e(e),
        .f(f),
        .g(g),
        .h(h),

        .y(y)
    );


    // ========================================
    // Clock generation
    // ========================================

    initial begin
        clk = 1'b0;

        forever #5 clk = ~clk;
    end


    // ========================================
    // Test
    // ========================================

    initial begin

        // ---------------------------------
        // Input values
        // ---------------------------------

        a = 16'd1;
        b = 16'd2;

        c = 16'd3;
        d = 16'd4;

        e = 16'd5;
        f = 16'd6;

        g = 16'd7;
        h = 16'd8;
```

```
        // ---------------------------------------
        // Reset
        // ---------------------------------------

        reset = 1'b1;

        #12;

        reset = 1'b0;


        // ---------------------------------------
        // Wait for 5-cycle schedule
        // ---------------------------------------

        #60;


        // ---------------------------------------
        // Display result
        // ---------------------------------------

        $display("");
        $display("=======================================");
        $display("          HLS DATAPATH TEST");
        $display("=======================================");

        $display("a = %0d", a);
        $display("b = %0d", b);
        $display("c = %0d", c);
        $display("d = %0d", d);
        $display("e = %0d", e);
        $display("f = %0d", f);
        $display("g = %0d", g);
        $display("h = %0d", h);

        $display("---------------------------------------");

        $display(
            "Expected y = %0d",
            ((a + b) * (c + d)) + (e + f) + (g * h)
        );

        $display(
            "Actual y   = %0d",
            y
        );

        $display("---------------------------------------");
```

```
        if (
            y ==
            ((a + b) * (c + d))
            + (e + f)
            + (g * h)
        ) begin

            $display("RESULT: PASS");

        end
        else begin

            $display("RESULT: FAIL");

        end


        $display("=====================================");

        $finish;

    end


    // ========================================
    // Waveform
    // ========================================

    initial begin

        $dumpfile("output/datapath.vcd");

        $dumpvars(0, tb_datapath);

    end

endmodule
```

This is the verification environment for the generated datapath.
It provides:

- Clock generation
- Reset
- Input stimulus
- Expected output calculation
- Actual output checking
- PASS/FAIL indication
- VCD waveform generation

For the example:

$$y = \big((1 + 2)(3 + 4)\big) + (5 + 6) + (7 \times 8)$$

so the testbench calculates the expected result and compares it with the generated hardware output.

Purpose**:** Verifies the functional correctness of the automatically generated RTL.

15. output/datapath.v

```
`timescale 1ns/1ps

module generated_datapath #(
    parameter WIDTH = 16
)(
    input  logic [WIDTH-1:0] a,
    input  logic [WIDTH-1:0] b,
    input  logic [WIDTH-1:0] c,
    input  logic [WIDTH-1:0] d,
    input  logic [WIDTH-1:0] e,
    input  logic [WIDTH-1:0] f,
    input  logic [WIDTH-1:0] g,
    input  logic [WIDTH-1:0] h,
    input  logic clk,
    input  logic reset,
    output logic [WIDTH-1:0] y
);

    // ============================================
    // Physical Registers
    // ============================================
    logic [WIDTH-1:0] R1;
    logic [WIDTH-1:0] R2;
    logic [WIDTH-1:0] R3;

    // ============================================
    // Control / Cycle Counter
    // ============================================
    logic [2:0] cycle;

    always_ff @(posedge clk) begin
        if (reset)
            cycle <= 3'd1;
        else if (cycle < 3'd5)
            cycle <= cycle + 3'd1;
        else
            cycle <= 3'd5;
    end

    // ============================================
    // MUX Outputs
    // ============================================
    logic [WIDTH-1:0] add_a;
    logic [WIDTH-1:0] add_b;
    logic [WIDTH-1:0] mul_a;
    logic [WIDTH-1:0] mul_b;

    logic [WIDTH-1:0] add_result;
```

```
    logic [WIDTH-1:0] mul_result;

    // ADD1 input A MUX
    always_comb begin
        case (cycle)
            3'd1: add_a = a;
            3'd2: add_a = c;
            3'd3: add_a = e;
            3'd4: add_a = R1;
            3'd5: add_a = R1;
            default: add_a = '0;
        endcase
    end

    // ADD1 input B MUX
    always_comb begin
        case (cycle)
            3'd1: add_b = b;
            3'd2: add_b = d;
            3'd3: add_b = f;
            3'd4: add_b = R3;
            3'd5: add_b = R2;
            default: add_b = '0;
        endcase
    end

    // MUL1 input A MUX
    always_comb begin
        case (cycle)
            3'd1: mul_a = g;
            3'd3: mul_a = R1;
            default: mul_a = '0;
        endcase
    end

    // MUL1 input B MUX
    always_comb begin
        case (cycle)
            3'd1: mul_b = h;
            3'd3: mul_b = R3;
            default: mul_b = '0;
        endcase
    end

    // ================================================
    // Functional Units
    // ================================================
    assign add_result = add_a + add_b;
    assign mul_result = mul_a * mul_b;

    // ================================================
```

```
    // Register Write / Output Logic
    // ==============================================
    always_ff @(posedge clk) begin
        if (reset) begin
            R1 <= '0;
            R2 <= '0;
            R3 <= '0;
            y <= '0;
        end
        else begin
            if (cycle == 3'd1) begin
                R1 <= add_result;
                R2 <= mul_result;
            end
            else if (cycle == 3'd2) begin
                R3 <= add_result;
            end
            else if (cycle == 3'd3) begin
                R1 <= mul_result;
                R3 <= add_result;
            end
            else if (cycle == 3'd4) begin
                R1 <= add_result;
            end
            else if (cycle == 3'd5) begin
                y <= add_result;
            end
        end
    end

endmodule
```

This is not manually written RTL. It is the final Verilog generated by verilog_generator.py using command:

- `iverilog -g2012 -o output/sim output/datapath.v tb/tb_datapath.sv`

It represents the hardware architecture produced by your HLS tool.
Purpose: Final RTL implementation of the scheduled and bound datapath.

16. output/datapath.vcd

```
$date
      Tue Sep 22 19:26:50 2026
$end
$version
      Icarus Verilog
$end
$timescale
      1ps
$end
$scope module tb_datapath $end
$var wire 16 ! y [15:0] $end
$var parameter 32 " WIDTH $end
$var reg 16 # a [15:0] $end
```

```
$var reg 16 $ b [15:0] $end
$var reg 16 % c [15:0] $end
$var reg 1 & clk $end
$var reg 16 ' d [15:0] $end
$var reg 16 ( e [15:0] $end
$var reg 16 ) f [15:0] $end
$var reg 16 * g [15:0] $end
$var reg 16 + h [15:0] $end
$var reg 1 , reset $end
$scope module dut $end
$var wire 16 - a [15:0] $end
$var wire 16 . b [15:0] $end
$var wire 16 / c [15:0] $end
$var wire 1 & clk $end
$var wire 16 0 d [15:0] $end
$var wire 16 1 e [15:0] $end
$var wire 16 2 f [15:0] $end
$var wire 16 3 g [15:0] $end
$var wire 16 4 h [15:0] $end
$var wire 1 , reset $end
$var wire 16 5 mul_result [15:0] $end
$var wire 16 6 add_result [15:0] $end
$var parameter 32 7 WIDTH $end
$var reg 16 8 R1 [15:0] $end
$var reg 16 9 R2 [15:0] $end
$var reg 16 : R3 [15:0] $end
$var reg 16 ; add_a [15:0] $end
$var reg 16 < add_b [15:0] $end
$var reg 3 = cycle [2:0] $end
$var reg 16 > mul_a [15:0] $end
$var reg 16 ? mul_b [15:0] $end
$var reg 16 @ y [15:0] $end
$upscope $end
$upscope $end
$enddefinitions $end
$comment Show the parameter values. $end
$dumpall
b10000 7
b10000 "
$end
#0
$dumpvars
bx @
b0 ?
b0 >
bx =
b0 <
b0 ;
bx :
bx 9
bx 8
```

```
b0 6
b0 5
b1000 4
b111 3
b110 2
b101 1
b100 0
b11 /
b10 .
b1 -
1,
b1000 +
b111 *
b110 )
b101 (
b100 '
0&
b11 %
b10 $
b1 #
bx !
$end
#5000
b111000 5
b1000 ?
b10 <
b111 >
b11 6
b1 ;
b1 =
b0 !
b0 @
b0 :
b0 9
b0 8
1&
#10000
0&
#12000
0,
#15000
b0 ?
b0 5
b0 >
b100 <
b111 6
b11 ;
b111000 9
b11 8
b10 =
1&
```

```
#20000
0&
#25000
b10101 5
b11 >
b101 ;
b111 ?
b1011 6
b110 <
b11 =
b111 :
1&
#30000
0&
#35000
b0 ?
b0 5
b0 >
b1011 <
b100000 6
b10101 ;
b1011 :
b10101 8
b100 =
1&
#40000
0&
#45000
b111000 <
b1011000 6
b100000 ;
b101 =
b100000 8
1&
#50000
0&
#55000
b1011000 !
b1011000 @
1&
#60000
0&
#65000
1&
#70000
0&
#72000
```

This is the waveform generated during RTL simulation.
It allows you to observe:

1. Clock

2. Reset
3. Cycle counter
4. MUX selections
5. Functional-unit outputs
6. Registers
7. Final output y

Purpose**:** Provides visual evidence that the generated datapath performs the intended computation correctly over the scheduled clock cycles.

Results:
1. for terminal commands:

- `python main.py`

```
PS C:\Users\Lenovo\EE677 Assignment 2> python main.py

========== DFG ==========

Primary Inputs:
['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h']

Operations:
N1: ADD ['a', 'b'] -> x1
N2: ADD ['c', 'd'] -> x2
N3: MUL ['x1', 'x2'] -> x3
N4: ADD ['e', 'f'] -> x4
N5: ADD ['x3', 'x4'] -> x5
N6: MUL ['g', 'h'] -> x6
N7: ADD ['x5', 'x6'] -> y

Primary Outputs:
['y']
=========================


========== DEPENDENCY GRAPH ==========

Node: N1
  Operation    : ADD
  Inputs       : ['a', 'b']
  Output       : x1
  Predecessors : []
  Successors   : ['N3']

Node: N2
  Operation    : ADD
  Inputs       : ['c', 'd']
  Output       : x2
  Predecessors : []
  Successors   : ['N3']

Node: N3
  Operation    : MUL
  Inputs       : ['x1', 'x2']
  Output       : x3
  Predecessors : ['N1', 'N2']
  Successors   : ['N5']

Node: N4
  Operation    : ADD
  Inputs       : ['e', 'f']
  Output       : x4
  Predecessors : []
  Successors   : ['N5']

Node: N5
  Operation    : ADD
  Inputs       : ['x3', 'x4']
  Output       : x5
  Predecessors : ['N3', 'N4']
  Successors   : ['N7']
```

```
Node: N6
  Operation    : MUL
  Inputs       : ['g', 'h']
  Output       : x6
  Predecessors : []
  Successors   : ['N7']

Node: N7
  Operation    : ADD
  Inputs       : ['x5', 'x6']
  Output       : y
  Predecessors : ['N5', 'N6']
  Successors   : []

====================================

========== TOPOLOGICAL ORDER ==========
N1 -> N2 -> N4 -> N6 -> N3 -> N5 -> N7
========================================

========== ASAP SCHEDULE ==========
Cycle 1: N1, N2, N4, N6
Cycle 2: N3
Cycle 3: N5
Cycle 4: N7
==================================


Overall schedule latency: 4 cycles

========== ASAP / ALAP / MOBILITY ==========
Node    Operation   ASAP    ALAP    Mobility
---------------------------------------------
N1      ADD         1       1       0
N2      ADD         1       1       0
N3      MUL         2       2       0
N4      ADD         1       2       1
N5      ADD         3       3       0
N6      MUL         1       3       2
N7      ADD         4       4       0
=============================================

Resource constraints:
  MUL: 1
  ADD: 1
  SUB: 1
  DIV: 1

========== LIST SCHEDULE ==========
Cycle 1: N1, N6
Cycle 2: N2
Cycle 3: N3, N4
Cycle 4: N5
Cycle 5: N7
==================================

========== OPERATION TIMES ==========
N1: Cycle 1
N2: Cycle 2
N3: Cycle 3
N4: Cycle 3
N5: Cycle 4
N6: Cycle 1
N7: Cycle 5
====================================
```

```
========== FUNCTIONAL UNIT BINDING ==========
Cycle   Node    Operation   Functional Unit
-----------------------------------------------
1       N1      ADD         ADD1
1       N6      MUL         MUL1
2       N2      ADD         ADD1
3       N3      MUL         MUL1
3       N4      ADD         ADD1
4       N5      ADD         ADD1
5       N7      ADD         ADD1
===============================================

Physical Functional Units:
  ADD1: N1, N2, N4, N5, N7
  MUL1: N6, N3
===============================================


========== REGISTER BINDING ==========
Variable    Live Range     Register
-------------------------------------
x1          [1, 3)         R1
x6          [1, 5)         R2
x2          [2, 3)         R3
x3          [3, 4)         R1
x4          [3, 4)         R3
x5          [4, 5)         R1
=====================================

Physical Registers:
  R1: x1, x3, x5
  R2: x6
  R3: x2, x4
=====================================

========== DATAPATH ==========

Inputs:
  a, b, c, d, e, f, g, h

Functional Units:
  ADD1
  MUL1

Registers:
  R1
  R2
  R3

Operations:
Cycle   Node    Operation FU      Inputs              Destination
-----------------------------------------------------------------
1       N1      ADD       ADD1    a, b                R1
1       N6      MUL       MUL1    g, h                R2
2       N2      ADD       ADD1    c, d                R3
3       N3      MUL       MUL1    x1, x2              R1
3       N4      ADD       ADD1    e, f                R3
4       N5      ADD       ADD1    x3, x4              R1
5       N7      ADD       ADD1    x5, x6              OUTPUT
=================================================================
```

```
========== MUX / CONTROL SIGNALS ==========
Cycle   Node    FU      Input A     Input B     Output
-----------------------------------------------------------
1       N1      ADD1    a           b           x1
1       N6      MUL1    g           h           x6
2       N2      ADD1    c           d           x2
3       N3      MUL1    R1          R3          x3
3       N4      ADD1    e           f           x4
4       N5      ADD1    R1          R3          x5
5       N7      ADD1    R1          R2          y
===========================================================

========== GENERATED NETLIST ==========

PRIMARY INPUTS:
  INPUT a
  INPUT b
  INPUT c
  INPUT d
  INPUT e
  INPUT f
  INPUT g
  INPUT h

REGISTERS:
  REGISTER R1
  REGISTER R2
  REGISTER R3

FUNCTIONAL UNITS:
  ADD1
  MUL1

MULTIPLEXERS:
  ADD1_A_MUX: R1, a, c, e
  ADD1_B_MUX: R2, R3, b, d, f
  MUL1_A_MUX: R1, g
  MUL1_B_MUX: R3, h

OPERATIONS / CONNECTIONS:
Cycle  Node   FU     A       B       DEST
-----------------------------------------------------------
1      N1     ADD1   a       b       R1
1      N6     MUL1   g       h       R2
2      N2     ADD1   c       d       R3
3      N3     MUL1   R1      R3      R1
3      N4     ADD1   e       f       R3
4      N5     ADD1   R1      R3      R1
5      N7     ADD1   R1      R2      OUTPUT

OUTPUTS:
  OUTPUT y
=======================================


=======================================
VERILOG NETLIST GENERATED
File: output/datapath.v
=======================================
```

```
========================================
GENERATING DFG VISUALIZATION
========================================

(process:6264): Pango-WARNING **: 18:26:20.998: Invalid UTF-8 string passed to pango_layout_set_text()

(process:22088): Pango-WARNING **: 18:26:21.182: Invalid UTF-8 string passed to pango_layout_set_text()

============================================
          DFG VISUALIZATION
============================================

Input file : input/example.dfg

Operations : 7
Inputs     : a, b, c, d, e, f, g, h
Outputs    : y

DOT file   : docs\dfg_auto.dot
PNG file   : docs\dfg_auto.png
SVG file   : docs\dfg_auto.svg

============================================

DFG VISUALIZATION GENERATED
Files:
  docs/dfg_auto.dot
  docs/dfg_auto.png
  docs/dfg_auto.svg
PS C:\Users\Lenovo\EE677 Assignment 2>
```

- iverilog -g2012 -o output/sim output/datapath.v tb/tb_datapath.sv
- vvp output/sim

```
PS C:\Users\Lenovo\EE677 Assignment 2> iverilog -g2012 -o output/sim output/datapath.v tb/tb_datapath.sv
PS C:\Users\Lenovo\EE677 Assignment 2> vvp output/sim
VCD info: dumpfile output/datapath.vcd opened for output.

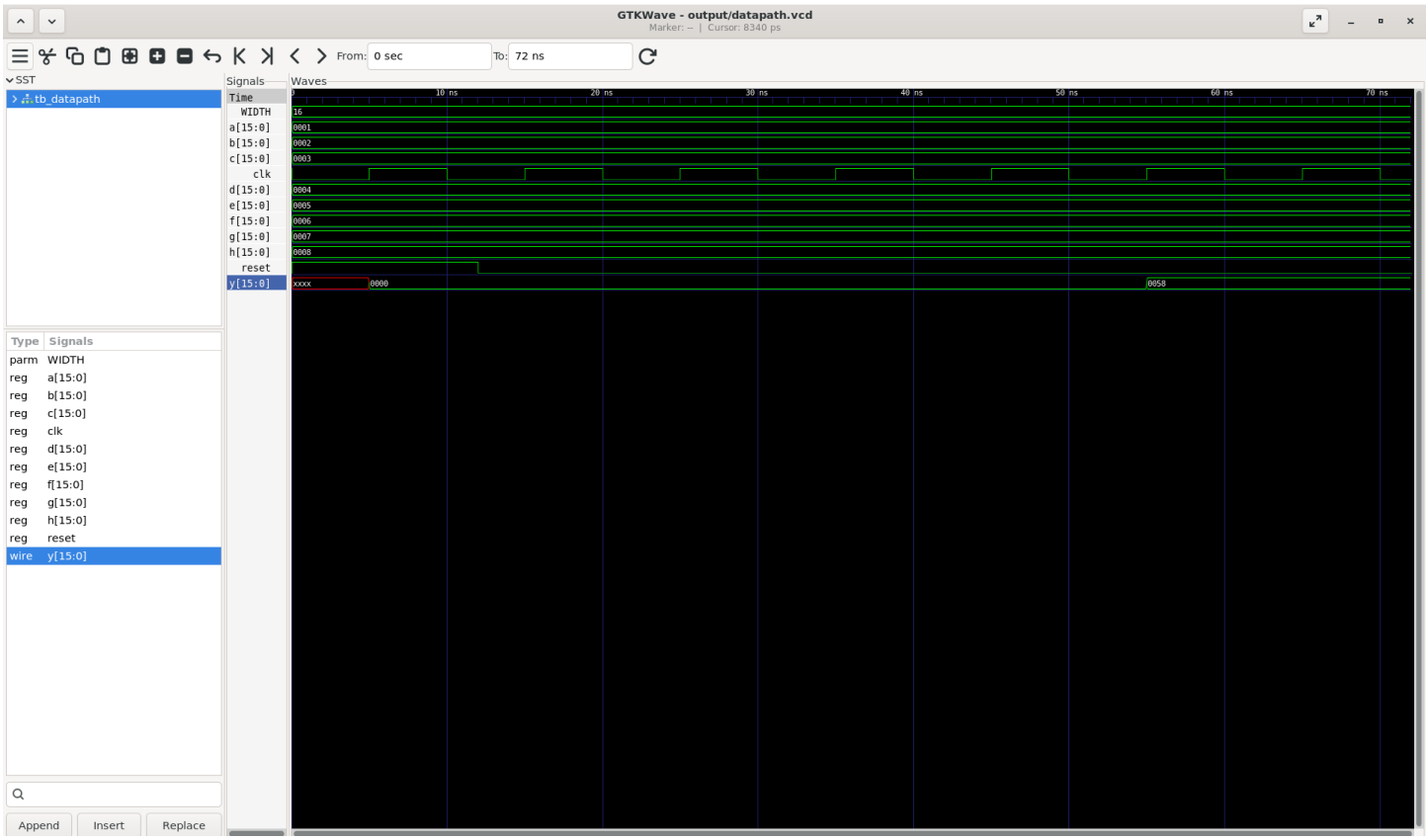
====================================
          HLS DATAPATH TEST
====================================
a = 1
b = 2
c = 3
d = 4
e = 5
f = 6
g = 7
h = 8
------------------------------------
Expected y = 88
Actual y   = 88
------------------------------------
RESULT: PASS
====================================
tb/tb_datapath.sv:150: $finish called at 72000 (1ps)
PS C:\Users\Lenovo\EE677 Assignment 2>
```

- gtkwave output/datapath.vcd

```
sureshvlsi@DESKTOP-CB90UTN:/mnt/c/Users/Lenovo/EE677 Assignment 2$ gtkwave output/datapath.vcd

GTKWave Analyzer v3.3.126 (w)1999-2026 BSI

[0] start time.
[72000] end time.
```



input values are:

Signal	Value
a	1
b	2
c	3
d	4
e	5
f	6
g	7
h	8

The HLS-generated schedule shown earlier was:

Cycle	Operation	Result
1	N1: a+b	3
1	N6: g×h	56
1	N2: c+d	7
2	N3: R1×R3	21
3	N4: e+f	11
4	N5: R1+R3	32
5	N7: R1+R2	88

Therefore:

$$R1 = (1 + 2)(3 + 4) = 3 \times 7 = 21$$
$$R3 = 5 + 6 = 11$$
$$R1 = 21 + 11 = 32$$
$$R2 = 7 \times 8 = 56$$

Finally:

$$y = 32 + 56 = \boxed{88}$$

The waveform displays:
y = 0058
because:

$$0058_{16} = 88_{10}$$

So the generated RTL produces the correct expected output.

2. Input DFG: docs/dfg.png

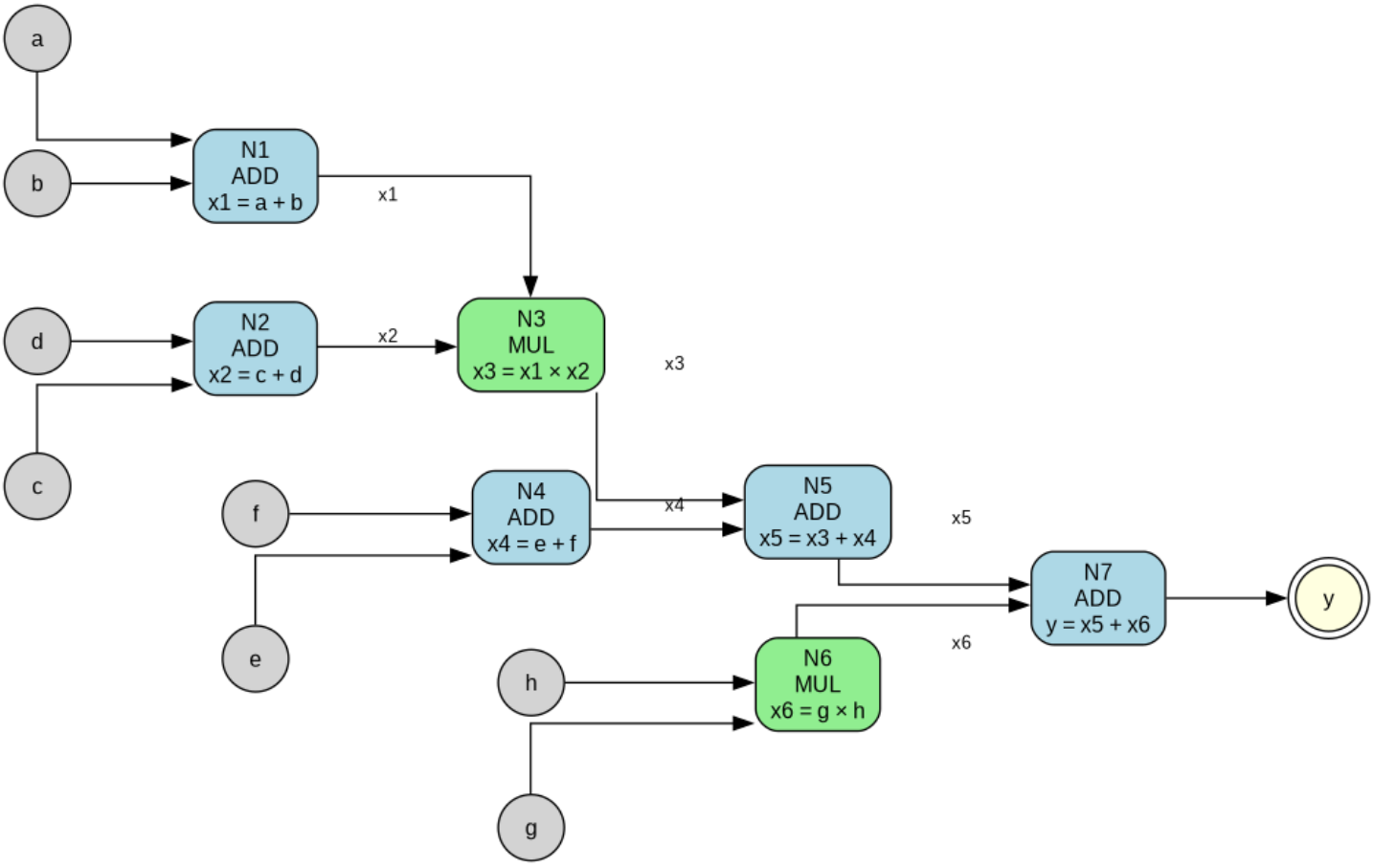


Figure: Input Data Flow Graph representing the computational specification.

3. Automatically Generated DFG: docs/dfg_auto.png

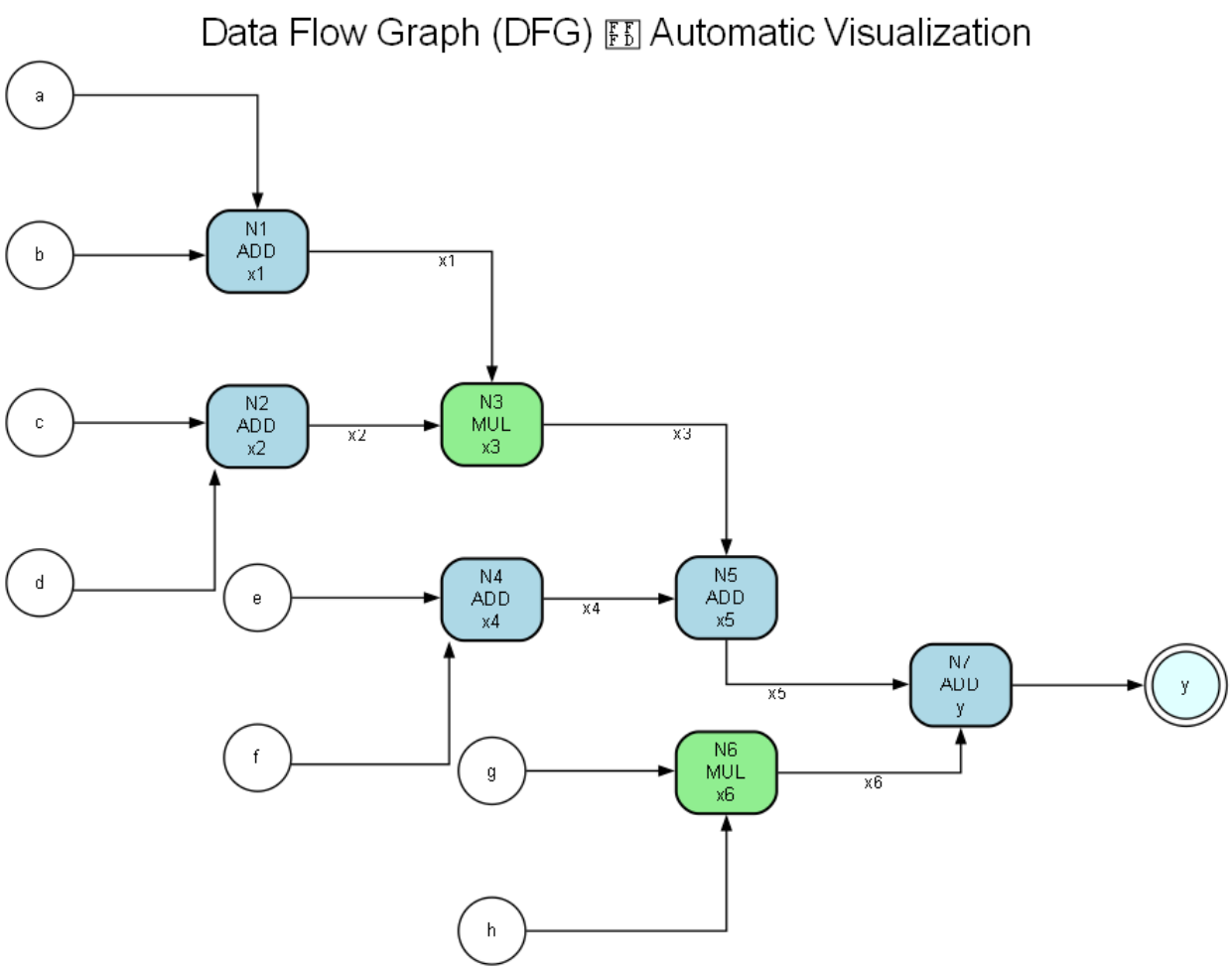


Figure: Automatically generated visualization of the input Data Flow Graph.

4. Scheduled DFG: docs/scheduled_dfg.png

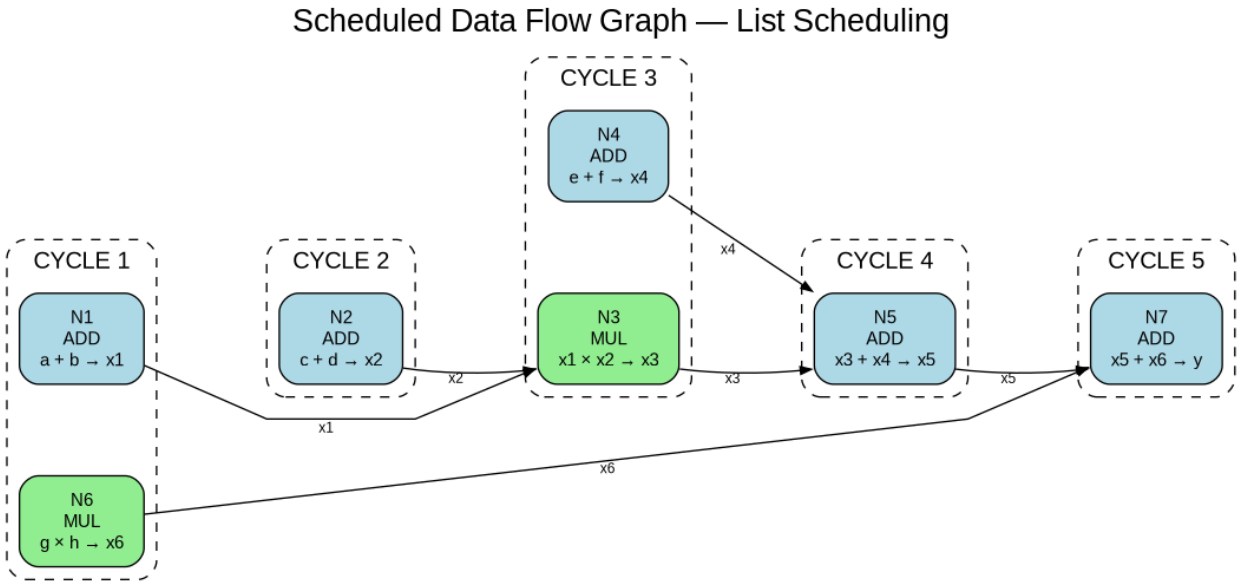


Figure: Scheduled DFG showing the assignment of operations to clock cycles.

Cycle	Node	Operation	Inputs	Destination
1	N1	ADD1	a, b	R1
1	N6	MUL1	g, h	R2
1	N2	ADD1	c, d	R3
2	N3	MUL1	R1, R3	R1
3	N4	ADD1	e, f	R3
4	N5	ADD1	R1, R3	R1
5	N7	ADD1	R1, R2	OUTPUT

5. Bound Datapath: docs/bound_datapath.png

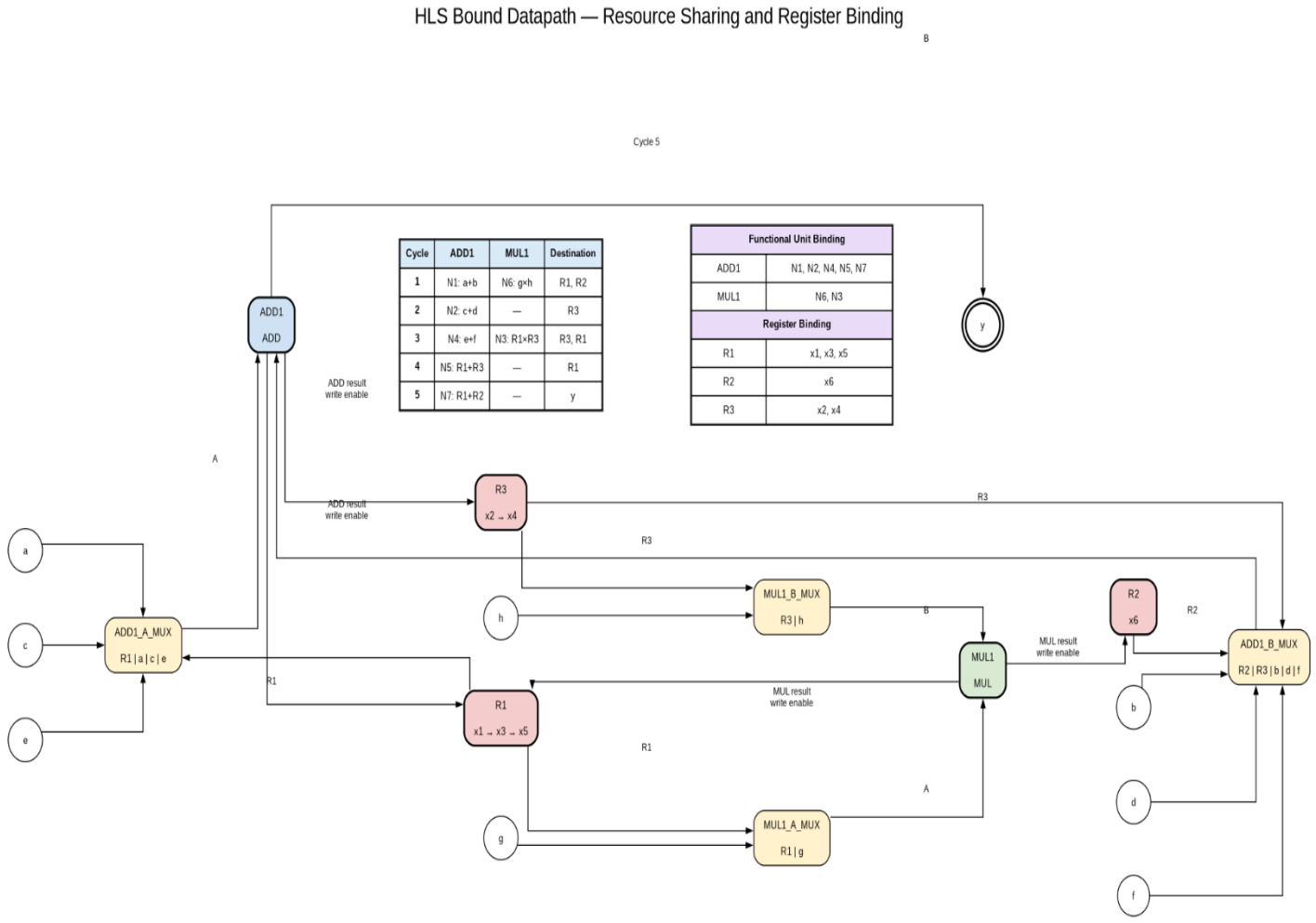


Figure: Bound datapath generated from the scheduled DFG after functional-unit and register binding.

Functional Unit Binding	
ADD1	N1, N2, N4, N5, N7
MUL1	N6, N3
Register Binding	
R1	x1, x3, x5
R2	x6
R3	x2, x4

6. Datapath Architecture: docs/datapath_architecture.png

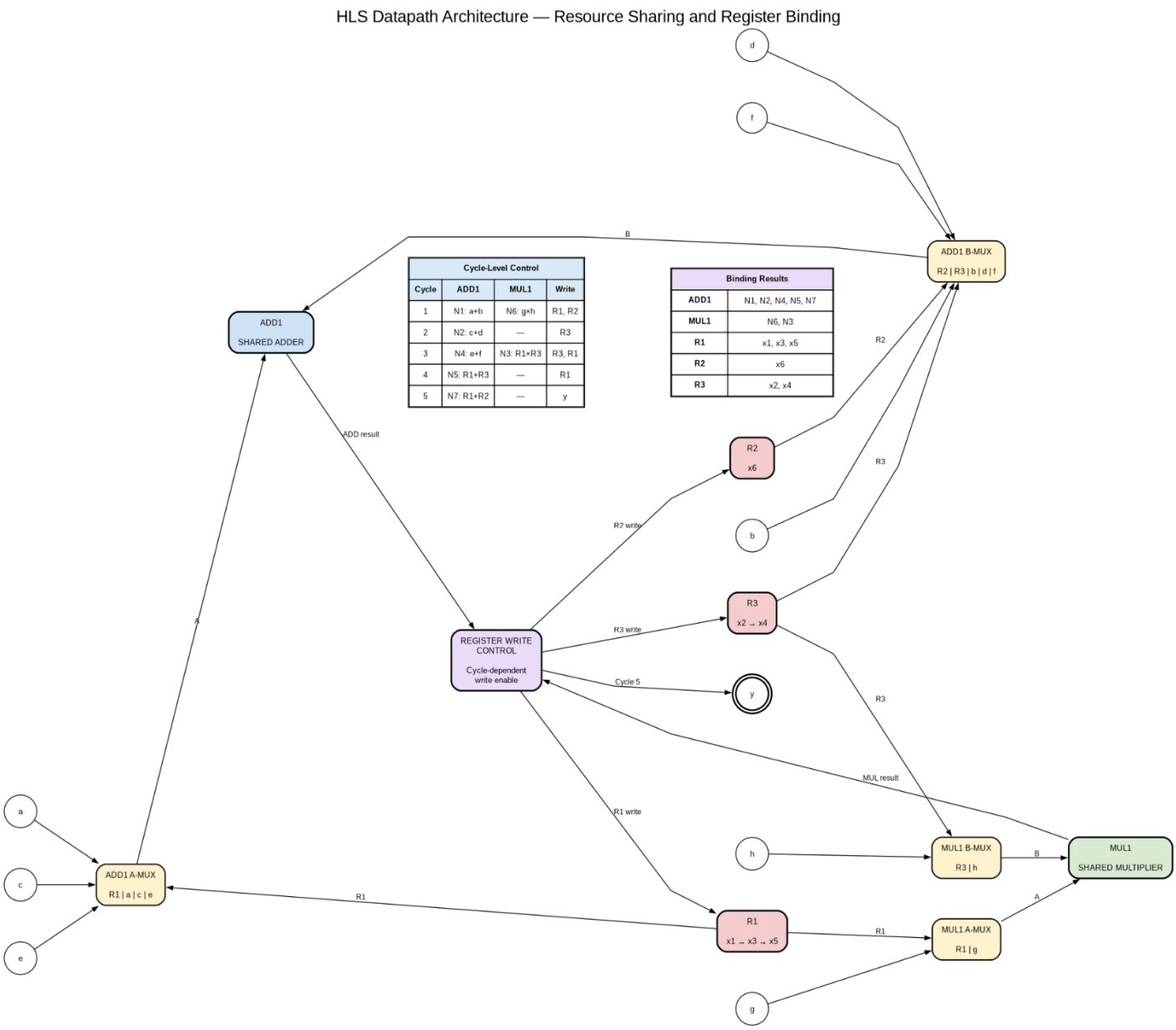


Figure: Generated datapath architecture consisting of functional units, registers, multiplexers, and interconnections.

Conclusion

This project presented the design and implementation of a Data Flow Graph (DFG)-based High-Level Synthesis framework for automatically transforming a behavioral computational specification into a hardware datapath and Verilog RTL implementation.

The developed framework accepts a DFG specification as input and performs the major stages of the HLS process, including DFG parsing, resource specification, scheduling, functional-unit binding, register binding, multiplexer and control generation, datapath construction, and RTL generation. The framework also provides graphical visualization of the DFG and scheduled DFG, enabling the intermediate stages of the synthesis process to be inspected and analyzed.

For the demonstrated example, seven computational operations were scheduled over five clock cycles while respecting data dependencies and the available functional resources. Functional-unit and register binding enabled hardware resources to be reused across different cycles, resulting in a structured datapath containing arithmetic units, registers, multiplexers, and associated control logic.

The generated Verilog RTL was successfully compiled using Icarus Verilog and functionally simulated using a dedicated testbench. The resulting VCD waveform was analyzed using GTKWave. For the test inputs $a = 1$, $b = 2$, $c = 3$, $d = 4$, $e = 5$, $f = 6$, $g = 7$, and $h = 8$, the generated hardware produced the final output $y = 88$. The waveform displayed the result as 0058 in hexadecimal, which corresponds to 88 in decimal, confirming the functional correctness of the generated RTL.

Overall, the project demonstrates the complete transformation from a high-level DFG representation to a verified RTL datapath. It provides a practical understanding of the fundamental HLS stages and illustrates how scheduling and resource binding influence the resulting hardware architecture. The framework can serve as a foundation for further work involving more advanced scheduling algorithms, resource-constrained optimization, latency-aware synthesis, power-aware optimization, automatic synthesis-report generation, and integration with commercial or open-source RTL synthesis and physical-design tools.

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